

BMS COLLEGE OF ENGINEERING



Flappy Bird Game on the ESP32

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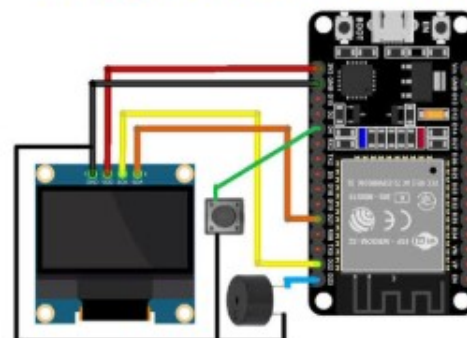
INTRODUCTION

Flappy Bird is a highly addictive and simple game where players control a bird that must navigate through a series of pipes without crashing. The game is known for its straightforward mechanics and challenging gameplay, requiring precise timing to keep the bird airborne. Inspired by its popularity, the project aims to recreate this classic game using the ESP32 microcontroller, integrating hardware components like an OLED display and buttons to replicate the original gaming experience. This setup not only provides a hands-on learning experience with microcontroller programming but also brings the nostalgia of Flappy Bird to life with modern electronics.

COMPONENTS

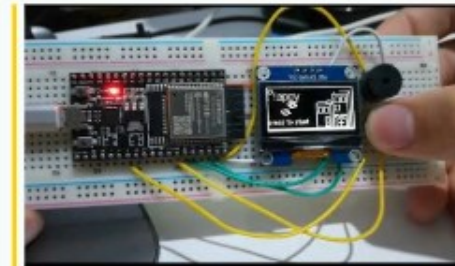
- ❑ ESP32 Microcontroller.
- ❑ OLED Display.
- ❑ Breadboard and Jumper Wires.
- ❑ Resistors.
- ❑ Push Button.
- ❑ Buzzer.
- ❑ Software Tools : Arduino IDE
- ❑ Cables and Connectors

Flappy Bird Game Circuit



CONNECTIONS

To set up the hardware, the OLED display is connected to the ESP32 using the I2C protocol, with the display's SDA and SCL pins connected to the corresponding I2C pins on the ESP32 (typically GPIO 21 for SDA and GPIO 22 for SCL). A button is connected to one of the GPIO pins on the ESP32, configured as an input. When the button is pressed, it sends a signal to the ESP32 to make the bird "flap" upwards. The ESP32 is powered through a USB connection or an external power supply, providing the necessary voltage to operate the microcontroller and the connected components. The entire setup is compact and efficient, making it ideal for portable gaming projects.



CHALLENGES AND SOLUTIONS

Creating the Flappy Bird game on the ESP32 involved ensuring smooth animation, responsive controls, and managing the timing and physics of the bird's movement. We optimized the game loop for better timing, implemented debounce logic for the button input, and used random number generation to place pipes at varying distances, ensuring smooth performance and engaging gameplay.

CONCLUSION

The Flappy Bird game on the ESP32 successfully combines hardware and software skills to recreate the classic game. This project highlights the capabilities of the ESP32 and offers a fun, educational experience in microcontroller programming and game development.